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Distributed framework for #SAT solvers

Thanh Phat Doan

Under the supervision of

Mihaela Sighireanu

Formal Methods Laboratory (LMF)

ENS Paris-Saclay

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Abstract

Exact model counting (#SAT) is a fundamental problem in automated reasoning with applications in verification, diagnosis, and probabilistic inference. Although sequential model counters have become highly effective, many instances remain too expensive to solve on a single machine. This report presents a distributed framework for exact model counting based on a master-worker architecture. The framework partitions the search space into disjoint instances of the input problem (called “cubes”), assigns them to workers, and aggregates the resulting counts. To reduce load imbalance caused by heterogeneous subproblem difficulty, it combines adaptive timeout estimation with dynamic cube resplitting. The implementation is solver-agnostic and uses gRPC for communication between components. We evaluate the framework on standard benchmark instances and discuss its limitations and possible extensions.

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1. Context

The Boolean Satisfiability Problem (**SAT**) represents the foundational challenge in automated reasoning, seeking to determine if there exists an assignment of truth values to variables that renders a given propositional formula true. These formulas are typically expressed in Conjunctive Normal Form (**CNF**), which consists of a conjunction of clauses, each representing a disjunction of literals. Consider the formula in CNF form $F = (x1 \vee x2) \wedge (\neg x1 \vee x3) \wedge (\neg x2 \vee \neg x3)$. A SAT solver aims to identify a satisfying assignment, such as $\{x1 = \top, x2 = \perp, x3 = \top\}$, which makes the formula evaluate to true.

This assessment of satisfiability is crucial across various domains, including hardware and software verification[1]. For instance, verifying that a software program adheres to its specifications can be translated into a SAT problem, where the program’s behavior is encoded as a propositional formula. Because of such wide-ranging applications, despite the NP-completeness of SAT (**Cook’s theorem** [2]), there has been significant progress in developing efficient solvers (notably CaDiCaL solver [3]) that can handle large and complex instances using sophisticated algorithmic techniques like Conflict-Driven Clause Learning (CDCL), chronological and non-chronological backtracking [4] to efficiently navigate and reduce the search space.

While SAT addresses the existence of a solution, Model Counting (**#SAT**) extends this problem by seeking to determine the total number of satisfying assignments within the 2^n possible configurations of a formula with n variables. This shift from decision to quantification elevates the problem to #P-complete [5], introducing a significantly higher level of computational complexity. However, #SAT enables a deeper analysis than classical SAT, providing insights into the solution space’s structure and size. Particularly in the case of software verification [6], the number of satisfying assignments can be used to estimate the proportion of inputs for which the program behaves correctly, providing a probabilistic measure of correctness. Another prominent application of #SAT is in security analysis, where it can be used to evaluate the robustness of cryptographic protocols [7] by not only determining if a vulnerability exists but also counting the number of ways an adversary can compromise a system and therefore assessing the risk level.

2. Motivation

Despite the advancements in current state-of-the-art #SAT solvers, many real-world instances remain computationally impractical for single-machine solving due to the exponential growth of the search space. Therefore, there is a pressing need for exploration of parallelization techniques in order to satisfy the demand of large-scale industrial benchmarks (e.g., circuit verification with millions of variables) or time-sensitive applications (e.g., probabilistic inference in machine learning). Unfortunately, as more complex algorithms and heuristics are developed for model counting, it is not trivial to adapt them to a distributed setting since they often rely on intricate interactions between components (component caching, clause learning, etc).

However, recent work has shown that model counting does exhibit parallel characteristics when the search space is appropriately decomposed. The cube-and-conquer paradigm [8] involves partitioning the search space into smaller “cubes” (partial assignments) and solving each cube independently, which can be effectively parallelized across multiple workers as demonstrated in [9]. This approach has shown significant speedups compared to sequential solving. However, [9] has only explored static strategies for cube generation and distribution, especially with adaptive strategies for handling heavy-tailed distributions of subproblem difficulty, where some subproblems may be solved very quickly, while others may require

significantly more time.

Accurately estimating the computational difficulty of these subproblems before solving them is challenging, and static partitioning strategies may lead to load imbalance and resource underutilization. Therefore, to address these problems, we propose the implementation of adaptive timeout mechanisms and dynamic work redistribution strategies. Furthermore, we intend to move beyond the constraints of traditional Message Passing Interface (**MPI**) by developing a flexible framework that supports heterogeneous deployment across mixed environments. By leveraging **gRPC** for inter-node communication, the solver can seamlessly integrate diverse resources, ranging from High-Performance Computing (HPC) clusters and cloud-based virtual machines to local workstations, into a unified distributed solving process. This modular architecture treats the underlying model counters as black boxes, allowing the framework to be easily adapted to any solver by simply wrapping its input and output interfaces within the distributed messaging layer.

3. Sequential #SAT solvers background

The efficiency of a distributed framework for #SAT solving is tied to the performance of its sequential backends. This chapter provides an overview of the current landscape of model counters and details the architecture of Ganak2, which serves as the primary engine for our framework.

3.1 Alternative solvers

The field of model counting has branched into several distinct algorithmic families. While our framework is designed to be solver-agnostic, that is it can accept any engine that follows the standard DIMACS CNF [10] format, understanding these alternatives is crucial for selecting the right backend for specific problem domains.

- **d4** : a knowledge compilation-based counter that compiles formulas into decision-DNNFs [11]
- **SharpSAT-TD** [12] : an evolution of the classic SharpSAT [13] that leverages **tree-decomposition** [14]. By identifying the structural width of the formula’s graph, it optimizes the order of component processing, often outperforming standard counters on industrial benchmarks with high regularity.
- **ApproxMC** [15] : an approximate model counter that provides probabilistic guarantees on the count. It is particularly useful for large instances where exact counting is infeasible or when an estimate is sufficient for the application at hand, offering a trade-off between accuracy and computational resources.

3.2 Ganak2

For the primary evaluation of this work, Ganak2 [16, 17] is chosen and utilized as the core sequential backend. Ganak2 is a state-of-the-art model counter that builds upon the sharpSAT architecture. Here’s a high-level overview of its architecture and key features:

Dual Independent Set Framework Utilizing the integrated **Arjun** [18] preprocessor, Ganak2 implements a dual-set strategy to balance search efficiency with SAT-solver transitions. It distinguishes between the **S-set** (SAT-eligibility set) and the **D-set** (Decision set):

- **S-set** ($S \subseteq P$): This is the minimal independent support identified by Arjun. Any two assignments agreeing on S define the values of all variables in the projection set P . Ganak2 uses this set as a threshold: once all variables in S are assigned, the solver can terminate the counting search and switch to a standard SAT solver to check the remaining formula.

- **D-set** ($D \supseteq P$): Unlike traditional counters that branch only on P , Ganak2 allows a larger decision set. By branching on variables outside the minimal support, the solver can more effectively decompose the formula into independent components, significantly increasing the “hit rate” of the component cache.

Formula Simplification Arjun employs two distinct phases to identify dependent variables: syntactic gate recovery (detecting AND and XOR gates) and implicit definability checks based on **Padoa’s Theorem** [19]. By leveraging an assumption-based CDCL framework, it prunes the formula of redundant variables before the search begins, preventing the solver from performing unnecessary computations on variables that are already logically defined by the support set.

Refined Residual Formula Processing Ganak2 optimizes the leaves of the search tree by switching to a modern SAT solver engine when the residual formula becomes simple enough. This engine incorporates state-of-the-art SAT heuristics such as VSIDS branching, **Luby-based restarts**, and **polarity caching**. Crucially, Ganak2 maintains a seamless state transition, allowing the solver to move between the counting phase and the SAT phase without losing learned clauses or forcing an expensive restart of the component cache.

The latest 2026 iteration of Ganak2 introduces native intra-node parallelism. Our framework invokes Ganak2 via the command line, typically using the following structure:

```
ganak2 formula.cnf --threads N
```

This creates a two-tier parallelization strategy: our framework manages inter-node distribution (across the cluster), while Ganak2 manages intra-node efficiency through multi-threading.

4. My contribution

This section presents the design and implementation of a distributed framework for exact model counting. The system is built around three principles. First, the search space is decomposed into independent cubes so that subproblems can be solved in parallel. Second, task execution is controlled through an adaptive timeout mechanism that reacts to the observed difficulty of the instance. Third, the framework remains solver-agnostic by treating the underlying model counter as an external backend accessed through a uniform interface.

The architecture follows a master-worker model. The master process is responsible for generating cubes, distributing tasks, collecting results, updating the global model count, and refining the timeout policy. Workers receive cubes, invoke a sequential solver on the corresponding restricted formulas, and report back either a model count or a timeout event. When a task exceeds the current deadline, the cube is split into smaller subcubes and returned to the work queue. This design allows the framework to adapt to instances with highly uneven subproblem difficulty while preserving the modularity needed to support different solvers and deployment environments.

The remainder of this chapter formalizes the cube decomposition strategy, describes the master and worker protocols, and explains the heuristics used for task redistribution.

4.1 Work partitioning via cube decomposition and dynamic load balancing

Cube decomposition Let F be a CNF formula over a set of variables X . A cube C is a conjunction of literals over a subset of X , that is,

$$C = \ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_k,$$

where each ℓ_i is a literal and no variable appears twice with conflicting polarity. The restricted counting problem associated with C is the formula $F \wedge C$.

A set of cubes $\mathcal{C} = \{C_1, \dots, C_m\}$ forms a partition of the search space if the cubes are pairwise contradictory and collectively exhaustive. In other words, for all $i \neq j$,

$$C_i \wedge C_j \equiv \perp,$$

and

$$\bigvee_{i=1}^m C_i \equiv \top$$

over the assignments that remain unspecified at the current level of decomposition. Under these conditions, exact model counting is additive:

$$\#(F) = \sum_{i=1}^m \#(F \wedge C_i).$$

This property makes cube decomposition particularly suitable for distributed solving, since each worker can process one subproblem independently and the final result is obtained by summing the partial counts.

In practice, however, the cost of solving the subproblems is not uniform. Some cubes are resolved quickly, while others lead to expensive search trees. This heavy-tailed behavior can create significant load imbalance if the decomposition is purely static. To address this issue, the framework uses adaptive timeout control together with dynamic resplitting of cubes that exceed their allotted time.

Adaptive timeout Let S denote the set of cubes that have already been solved successfully, and let $t(C)$ denote the solving time of cube C . The timeout threshold is updated from the observed runtime history:

$$T_{\text{limit}} = \alpha \cdot \frac{1}{|S|} \sum_{C_i \in S} t(C_i),$$

where $\alpha > 0$ is a scaling factor. This policy allows the timeout to evolve with the actual difficulty of the benchmark rather than remaining fixed throughout the execution. At the beginning of a run, the timeout may be initialized to $+\infty$ until enough observations are available to compute a meaningful estimate.

Branching heuristics We define a heuristic function

$$h : (F, C) \mapsto (x_1, \dots, x_k), \quad 1 \leq k \leq k_{\text{max}},$$

which selects a small set of branching variables from the restricted formula $F \wedge C$. The parameter k_{max} is kept small (typically $k \in \{1, 2, 3\}$) to avoid excessive splitting which can lead to an explosion of new cubes in the queue.

The heuristic can be instantiated in several ways depending on the desired trade-off between computational cost and decomposition quality. For example, it may select variables uniformly at random,

prioritize variables with high clause occurrence, exploit structural properties in tree decompositions [14], or reuse solver-specific signals such as VSIDS scores. The goal is to identify variables whose assignment is likely to significantly simplify the remaining formula.

Cube splitting function Given $(x_1, \dots, x_k) = h(F, C)$, the cube C is refined by enumerating all assignments to the selected variables. For each vector $\sigma \in \{0, 1\}^k$, define:

$$\ell(\sigma) = \bigwedge_{i=1}^k \begin{cases} x_i & \text{if } \sigma_i = 1, \\ \neg x_i & \text{if } \sigma_i = 0. \end{cases}$$

The splitting function is then defined as:

$$f_{\text{split}}(F, C) = \{C \wedge \ell(\sigma) \mid \sigma \in \{0, 1\}^k\}.$$

This construction ensures that the resulting cubes form a disjoint partition of the original cube:

$$\bigvee_{C' \in f_{\text{split}}(F, C)} C' \equiv C, \quad C' \wedge C'' \equiv \perp \text{ for } C' \neq C''.$$

This formulation generalizes classical binary branching. When $k = 1$, the cube is split into two subcubes, whereas larger values of k produce 2^k subcubes in a single step. In contrast to static decomposition strategies, splitting is applied selectively only to cubes that are empirically hard, as indicated by timeout events. This avoids unnecessary overhead on easy subproblems while allowing finer-grained parallelism where it is most needed.

However, this redistribution approach faces a significant inherent challenge: the loss of accumulated work of restarting a solver on a new cube. When a worker abandons a cube due to a timeout, the progress made during that search is lost unless the solver’s internal state, such as cached components or learned clauses. While such recovery is technically feasible, it often requires dissecting the internal particularities of specific solvers, which is outside the scope of our solver-agnostic framework. To mitigate this lost work penalty, we aim to minimize timeout occurrences by carefully calibrating the granularity of the initial decomposition. By generating cubes that are sufficiently large to justify the overhead of a distributed start, yet fine-grained enough to ensure that the relative difficulty between subproblems remains balanced, we can reduce the frequency of re-splitting and maximize overall throughput.

4.2 Master-side

The framework follows a master-worker architecture, where a central coordinator (master) is responsible for generating the cubes (tasks) and distributing them to a pool of worker nodes. The master operates in a fully asynchronous, event-driven manner, where all interactions are triggered by **RPC** calls from workers. Task distribution is handled through the **GetTask** procedure: when a worker becomes idle, it requests a new cube from the master. If no tasks are available, it receives a special payload having *task_id* set to -1 , signaling it to wait for a bit and retry later.

Upon receiving a result via **SubmitResult**, the master distinguishes between completed and timed-out tasks. Successful executions contribute directly to the global model count and are recorded to update the dynamic timeout. The timeout T_{limit} is recomputed as a scaled average of previously observed solving times, allowing the system to adapt to the hardness of the instance over time. Updated timeout values are propagated to workers through a streaming RPC mechanism. It should be noted that the master doesn’t

Algorithm 1: Master

have to broadcast constantly the new updated timeout value if the change is not significant enough (e.g., less than 10% difference from the previous value) to avoid unnecessary communication overhead and network congestion.

In the case of a timeout, the corresponding cube is refined using the splitting function f_{split} guided by specific heuristics to generate a set of smaller subcubes. These are then reinserted into the queue, from which they will be distributed to workers as they become available.

The architecture naturally supports scalability as the worker pool is not fixed and can be extended dynamically as new workers establish gRPC connections to the master. Since workers independently pull tasks and report results, the system achieves load balancing without prior knowledge of available computational resources.

4.3 Worker-side

The worker follows a pull-based execution model, where tasks are requested from the master on demand via the `GetTask` RPC. This eliminates the need for centralized scheduling and allows workers to operate independently. Each task is executed by spawning an external solver process, which is invoked with the appropriate command-line arguments to solve the restricted formula $F' \equiv F \wedge C$. This process-level isolation enables safe termination and restart of the solver in response to timeout conditions.

To support adaptive execution, workers subscribe to a streaming RPC interface `SubscribeTimeoutUpdates` and continuously receive updated timeout values from the master. These updates are monitored asynchronously, and if a new timeout is exceeded during execution, the worker terminates the solver process early and reports the task as timed out.

The worker also emits periodic status updates containing runtime metrics such as CPU and memory usage. These updates are used solely for monitoring and visualization purposes and do not influence the solving process.

Finally, workers can join or leave the system at any time by establishing or closing a gRPC connection to the master. Once the master detects a new worker connection, it adds it to the pool of available workers and starts assigning tasks to it. If a formula has been solved, the master can still take in a new problem and start the process again with current workers without any disruption. Since each worker is fully decoupled from the others and manages its execution and communicates to the master only through RPC interfaces, the system remains robust and adaptable to varying computational environments.

5. Implementation and evaluation

5.1 Implementation

General Architecture Our framework is built upon a distributed master-worker architecture designed to handle large-scale parallelized model counting. The system leverages gRPC for high-performance communication, utilizing a pull-based task distribution model where workers request formulas and the master provides centralized scheduling and result aggregation. As illustrated in Figure 5.1, this design incorporates real-time telemetry via bidirectional streams and a dedicated TUI for live monitoring of the global solving state.

Communication layer The distributed components communicate through gRPC and Protocol Buffers. This choice provides a strongly typed interface between the master, the workers, and the terminal user

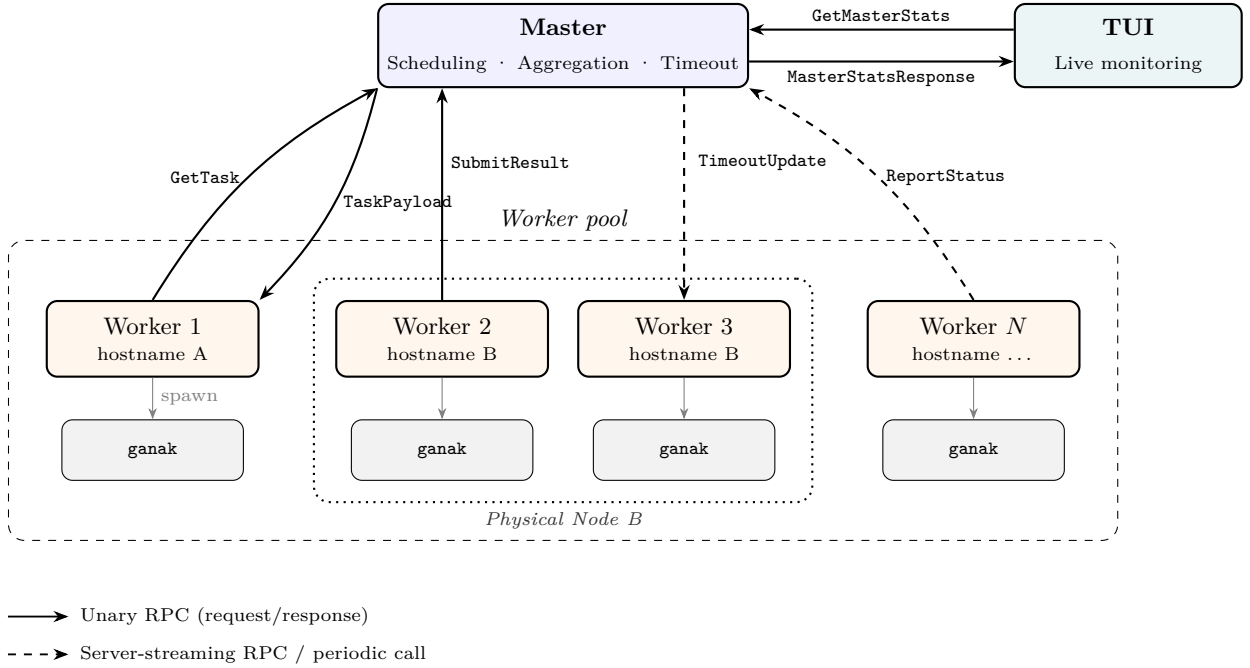


Figure 5.1: Distributed master-worker architecture.

interface. The protocol definitions are compiled into native client and server code, which reduces serialization errors and avoids the overhead of text-based formats. gRPC also supports bidirectional streaming, which is useful for continuous status updates during solving without blocking the main request-response flow. Since the system is intended to run across heterogeneous environments, the use of standard TCP/IP communication makes it possible to deploy workers on different machines, operating systems, and containerized environments without requiring a homogeneous cluster.

Master implementation The master is implemented in Go. Go is well suited to the coordination layer because its concurrency model makes it straightforward to manage many simultaneous worker connections. Goroutines and channels provide lightweight concurrency primitives, while the standard library offers direct support for gRPC, synchronization, cancellation, and arbitrary-precision arithmetic. In addition, Go produces a single statically linked binary, which simplifies deployment and makes the master process suitable for long-running server execution.

Worker implementation The workers are implemented in Python. This choice is motivated by the ease of process control, since the solver is executed as an external subprocess whose standard output, errors, and lifetime must be monitored and controlled. Python also provides convenient access to system-monitoring libraries and simplifies experimentation with different solver integrations. Because the worker logic mainly orchestrates execution rather than performing heavy computation itself, Python offers a practical balance between flexibility and implementation speed.

Containerization The system is packaged with Docker to ensure reproducibility and simplify deployment. Each worker can run inside an isolated container with the solver pre-installed, which avoids dependency conflicts and helps guarantee consistent execution environments. Docker Compose is used to orchestrate the master, the worker replicas, and the supporting services. This approach supports both fully containerized deployments and hybrid setups in which only part of the system is containerized.

Terminal user interface The terminal user interface is implemented in Go using Bubble Tea. Its architecture follows the event-driven style of the Elm framework: the model stores the current state of the system, updates react to events such as user input or gRPC responses, and the view renders the state to the terminal. This structure is appropriate for a monitoring interface because it keeps rendering separate from data collection and prevents the interface from blocking while waiting for network responses.

Build system A Makefile coordinates protocol generation, compilation, and deployment. It automates code generation from the Protocol Buffer definitions, builds the Go binaries, and launches the distributed environment with a single command. This reduces manual setup and ensures that the master, the workers, and the interface remain synchronized with the same communication schema.

5.2 Evaluation

6. Conclusion and future work

6.1 Conclusion

Our framework addresses the scalability limitations of sequential model counting through a modular, gRPC-based distributed framework designed for heterogeneous environments. By decoupling the coordination layer from the underlying sequential engines, the framework maintains a solver-agnostic architecture that can integrate any DIMACS-compatible tool, such as Ganak2 or d4. The primary contribution of this work lies in the implementation of adaptive mechanisms, specifically dynamic cube resplitting and a self-calibrating timeout strategy, to mitigate the performance bottlenecks caused by heavy-tailed distributions of subproblem difficulty. Preliminary results indicate that while the framework achieves significant speedups.

6.2 Limitations and improvements

Preliminary results indicate that the framework can reduce overall solving time on instances with uneven subproblem difficulty, although its effectiveness depends on the quality of the initial decomposition and the variability of the backend solver.

A primary area of improvement lies in the cube decomposition process, where the integration of structural heuristics such as tree decomposition or VSIDS-based [20](Variable State Independent Decaying Sum) variable selection could facilitate the generation of more balanced and higher-quality subproblems.

Furthermore, while the current dynamic timeout mechanism relies on a simple average of historical execution times, a more robust probabilistic model could be developed to estimate T_{limit} based on the structural similarities between active and previously solved cubes.

To further optimize resource allocation, a "scouting" mechanism could be implemented where a subset of workers utilizes fast SAT heuristics to gauge the complexity of subproblems that appear stagnant. This approach would allow the system to intelligently distinguish between truly intractable sub-cubes and those nearing completion, enabling more informed decisions on whether to trigger a redistribution or grant a "mercy" extension to the current worker.

Beyond heuristic refinements, long-term research will explore the integration of distributed hash tables or databases to facilitate component cache sharing across the network. This would require a more deep integration with the internal workings of the underlying solvers, but it has the potential to significantly reduce redundant computations by allowing workers to benefit from the progress of their peers.

Additionally, a portfolio-based approach could be implemented to deploy multiple solver types simultaneously, allowing the framework to automatically select the most efficient backend for a given sub-cube.

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